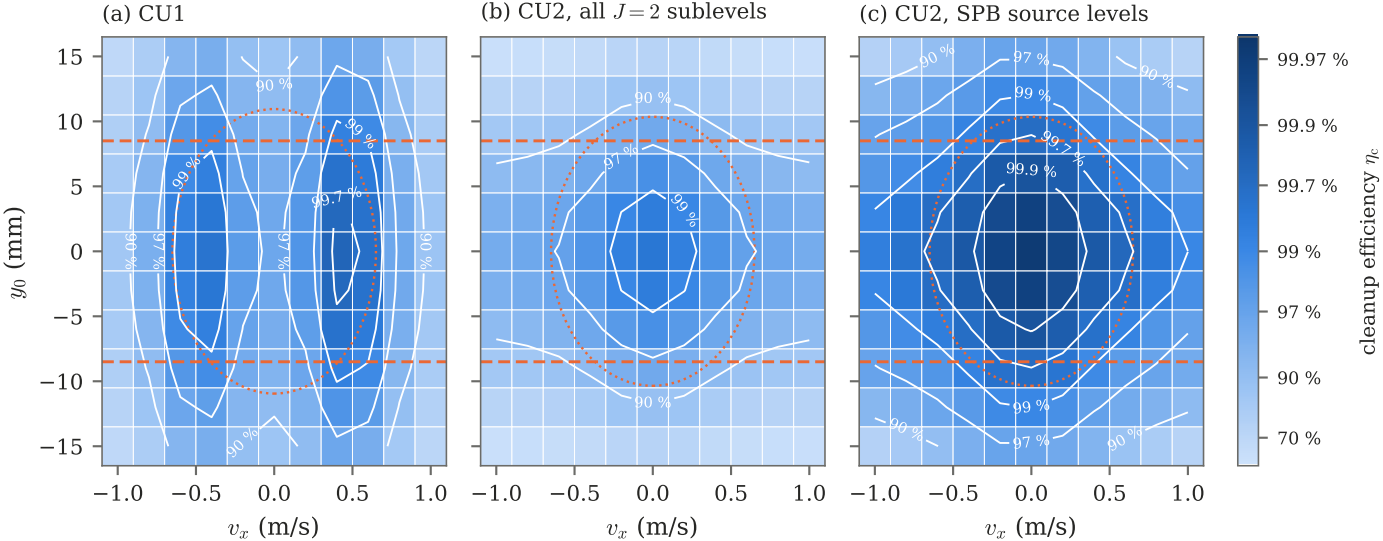


R(1) $F_1' = 5/2$, $F' = 2$; 150 V/cm
$\delta/2\pi = +19.0$ MHz, no EOM, 80 mW
beam 4.8 cm high ($\sigma_y = 12$ mm)
start: 8 SPB-related $J = 1$ levels
on axis: 96.36 %
mean in acceptance: **97.1** %

R(2) $F_1' = 7/2$, $F' = 3 + F' = 4$; 200 V/cm
setting R1: $f_{\mathrm{AOM}} = 208.5$ MHz,
$\delta_0/2\pi = +11.5$, $\delta_1/2\pi = +8.17$ MHz
80 + 24 mW; EOM 17.5 MHz, $\beta = 1.5$
beam 3 cm high ($\sigma_y = 7.5$ mm)
start: all 20 $J = 2$ sublevels
on axis: 99.52 %
mean in acceptance: **96.7** %

R(2) $F_1' = 7/2$, $F' = 3 + F' = 4$; 200 V/cm
setting R1: $f_{\mathrm{AOM}} = 208.5$ MHz,
$\delta_0/2\pi = +11.5$, $\delta_1/2\pi = +8.17$ MHz
80 + 24 mW; EOM 17.5 MHz, $\beta = 1.5$
beam 3 cm high ($\sigma_y = 7.5$ mm)
start: 4 $J = 2$, $m_J = 0$ levels
(lens weights 2/3, 1/9, 1/9, 1/9)
on axis: 99.97 %
mean in acceptance: **99.6** %

Dashed orange: detection acceptance $|y_0| \le 8.5$ mm (detection laser 1.7 cm high); the grid spans the accepted $|v_x| \le 1$ m/s.
Dotted orange: 1σ of the beam distribution at 25 kV lens voltage ($\sigma_{v_x} = 0.65$ m/s; $\sigma_y = 10.95$ mm at CU1, lens + 0.5 m; 10.35 mm at CU2, lens + 1.0 m), which also weights the mean in acceptance. White lines: contours of η_c. On axis: $v_x = 0$, $y_0 = 0$.
Both stages: 13 passes, 10 % loss per pass, X/Z switching at 0.5 MHz, Doppler shift $\pm k v_x$ alternating pass by pass; single trajectories at $v_z = 184$ m/s; grid 0.2 m/s × 3 mm, computed for $y_0 \ge 0$ and mirrored.